

Econ 30065: Microeconomics

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Externalities

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- Today, we see another reason for inefficiency: *externalities*
- One of the key assumption of the competitive market model we had was that
 - Each consumer's utility depends only on her own consumption
 - Each consumer's action affects only his own well-being
- In reality this assumption is often violated:
 - A person delighting neighbors playing piano
 - A power generator polluting air, contributing to respiratory diseases
 - Article we saw at the beginning on gas production
- The actions in these example create *externalities*

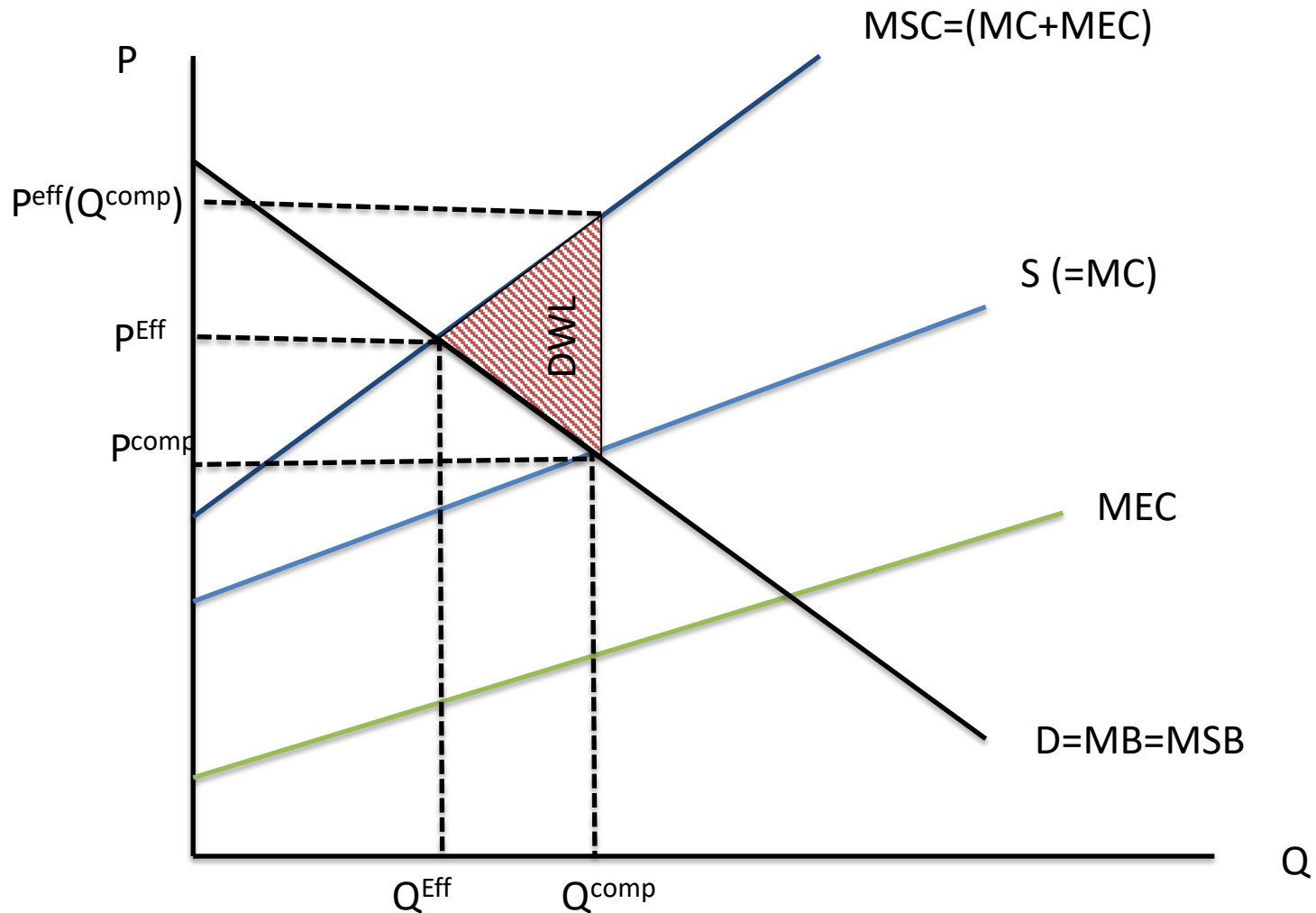
Externalities

- An action creates an ***externality*** if it affects someone with whom the decision-maker has not engaged in a related market transaction
 - ***Negative externality*** if it harms someone else
 - Polluting causes **health problems in a community**
 - ***Positive externality*** if it benefits someone else
 - Carefully tended garden causes a **raise in the value of surrounding homes**

Negative Externalities and inefficiency in competitive markets

- Presence of a negative externality in a competitive market will usually allocate economic resources inefficiently
 - With a negative externality there is an **external cost**, the economic harm that a negative externality imposes on others:
 $EC(Q) > 0$ ($MEC(Q) > 0$)
 - Optimality condition in competitive markets: each firm produces up to the point at which **its** marginal cost curve equals its marginal revenue (market price): $P = MC$
 - Since $P = MB \rightarrow MB = MC$
 - But for a level of production to be socially efficient it must be that marginal **social** cost and marginal **social** benefit are equal
 - Marginal social benefit is equal to marginal benefit (no positive externality)
 - Marginal social cost = marginal cost to the producer + marginal external cost
- $$MSC = MC + MEC$$
- Therefore the firm's equilibrium production level is socially excessive and inefficient: $MB = MSB < MSC$

Negative Externality in a Competitive Market



Negative Externality in a Competitive Market

	Perfect competition	Efficient level of output
CS	$A+B+C+D$	A
PS	$E+F+G$	$B+C+E+F$
External cost	$C+D+F+G+H$	$C+F$
Social welfare	$A+B+C+D+E+F+G-C-D-F-G-H=A+B+E-H$	$A+B+C+E+F-C-F=A+B+E$

